

Alex Pavel Medina Jorquera

MSC IN COMPUTER SCIENCE · MATHEMATICAL AND COMPUTATIONAL ENGINEERING · SRE

Santiago, Chile

☎ (+56 9) 9868 7017 | ✉ apavelmj@gmail.com | 📷 alichwan | 🌐 pavelmedinajorquera | 🏠 My site

Summary

MSc in Computer Science and Mathematical and Computational Engineer, with a background in data science and experience in software engineering, machine learning, and systems reliability. My quantitative training is complemented by experience developing and operating production systems, allowing me to approach problems from different angles and solve them primarily through data and software. Experience in Python, backend development, cloud infrastructure, distributed systems, natural language processing, and automation.

Experience

Zipedi

Santiago, Chile

SERVICE RELIABILITY ENGINEER (ON-CALL) · L0 → L1

Apr 2023 – Apr 2026

Internationally-operating software company running a distributed fleet of autonomous robots to build digital twins of retail stores, contrast their physical state against operational data, and generate tasks for on-site teams.

- Promoted from SRE L0 to SRE L1 in late 2023, as part of my progression within the team.
- Replaced a reactive workflow in which service failures were manually reported by Customer Success and Support to the development team, by implementing automated alerts that detected issues as they occurred and quickly routed them to the responsible team.
- To address the difficulty of simultaneously monitoring robots operating across different stores and locations, I developed task-latency-threshold-based monitoring mechanisms that continuously evaluated their behavior and generated alerts on anomalous conditions.
- Investigated software, infrastructure, hardware, and operational incidents through telemetry, logs, and system behavior, identifying root causes and recurring patterns.
- Developed and maintained microservices and internal tools in Python, deployed on GCP infrastructure, to support the operation and reliability of production systems.

CORFO — Corporación de Fomento de la Producción

Santiago, Chile

DATA SCIENCE INTERN (2020) · MACHINE LEARNING INTERN (2022)

2020 / 2022

Chilean public agency that promotes entrepreneurship, innovation, and competitiveness through multiple funding instruments, programs, and calls for proposals aimed at companies and projects.

- **Machine Learning Intern (2022).** To reduce the manual work required to standardize historical applications, I took part in developing a Python-based text classification system on data that included 9,000 awarded projects and 12,000 submitted applications.
- Evaluated text representations using BETO, multilingual embeddings, and Doc2Vec, together with classic classifiers such as SVM, KNN, decision trees, and neural networks.
- **Data Science Intern (2020).** To reduce manual review of potentially duplicate proposals, I designed similarity metrics and implemented vectorization and text processing pipelines in Python using Pandas, Scikit-learn, and natural language processing tools.

Corte Suprema de Chile / IMFD — Instituto Milenio Fundamentos de los Datos

Santiago, Chile

DEVELOPER INTERN

Jan 2021 – Mar 2021

Project commissioned by the Corte Suprema de Chile to the Instituto Milenio Fundamentos de los Datos to develop legal document analysis tools using machine learning.

- To decouple the configuration and execution of machine learning models from their internal implementation, I developed a REST API in Flask that allowed selecting, training, and running inference on different models for legal document analysis.

Education

Pontificia Universidad Católica de Chile

Santiago, Chile

MASTER'S IN ENGINEERING SCIENCES — COMPUTER SCIENCE

Mar 2022 – Jan 2025

Concentration: Machine Learning

Pontificia Universidad Católica de Chile

Santiago, Chile

DEGREE IN MATHEMATICAL AND COMPUTATIONAL ENGINEERING

Mar 2021 – Jan 2025

Major: Data Science | Minor: Uncertainty Quantification

Pontificia Universidad Católica de Chile

Santiago, Chile

BACHELOR'S IN ENGINEERING SCIENCES

Mar 2015 – Dec 2020

Research

Comparing MILP, SAT, and ASP for the Synthesis of Minimal Automata from Traces

MASTER'S THESIS

Aug 2021 – Jan 2025

- Evaluated and compared three formal approaches: Mixed Integer Linear Programming, Boolean Satisfiability (*SAT solvers*), and Answer Set Programming; to synthesize minimal deterministic finite automata from labeled example traces.

Synthesis of LTL Specifications from Traces

UNDERGRADUATE RESEARCH (IPRE)

2021

- Investigated methods to infer *Linear Temporal Logic* (LTL) specifications from traces (sequences of events, steps, or decisions), aiming to formally represent observed behavior through temporal constraints.

Inspecting the Concept Knowledge Graph Encoded by Modern Language Models

RESEARCH COLLABORATION

2021

- Ran semantic probing experiments to study whether the representations learned by language models encoded hierarchical semantic relations between WordNet concepts.
- The research later evolved into a systematic study of the semantic knowledge encoded by different language models; my contribution during its early stages was acknowledged by the authors in the publication's acknowledgments section **arXiv:2105.13471**.

Skills

Programming	Python, SQL, Bash, JavaScript, TypeScript
Data Science & ML	Pandas, NumPy, NLP, Scikit-learn, PyTorch, TensorFlow
Visualization	Matplotlib, Seaborn, Plotly Express
Backend & APIs	Flask, REST APIs, Microservices, FastAPI
Data Engineering	BigQuery, dbt, Airflow
Databases & Storage	PostgreSQL, Google Cloud Storage, MongoDB, Redis (basic)
Frontend	HTML/CSS, React
Cloud & DevOps	GCP, Cloud Run, Pub/Sub, Docker, Terraform, GKE
Development	Git, GitHub, GitLab
Logic Programming	Clingo/ASP
Languages	Spanish (Native), English (B2), Dutch (A1)